Regime-Switching Asset Allocation via a POMDP

A QMDP regime overlay that beats 60/40, stress-tested against eleven baselines

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Have you ever wondered...

Why does the "safe" 60/40 portfolio sometimes lose money when you most need it not to?

In **2022**, the textbook 60/40 stock-bond portfolio lost $\sim$**17%** as stocks AND bonds fell together.

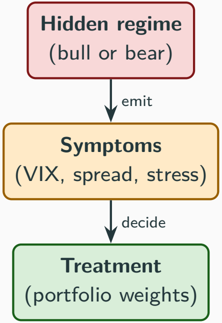
- Static allocation assumes joint return distribution is stable. Reality: conditional heteroskedasticity + stress correlation.
- Question. Can a fully decision-theoretic regime overlay beat 60/40 out-of-sample after realistic transaction costs?

The doctor analogy (= POMDP)

You are a doctor. Your patient is the market.

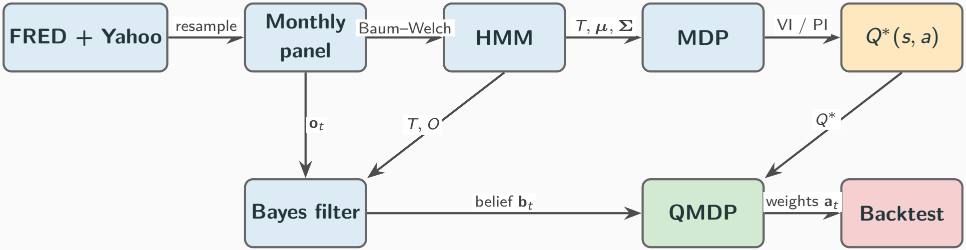
- Patient is *healthy* (bull) or *sick* (bear), but you can't examine directly.
- You only see symptoms (VIX, term spread, stress indices).
- Different treatments (portfolio weights) work better in different states.
- Goal: read symptoms, infer state, prescribe.

This IS a POMDP: s_t = regime, $\mathbf{o}_t$ = obs, $\mathbf{a}_t$ = weights, R = expected utility.



Tree, MDP, ID all fail here. POMDP is the minimum tool.

End-to-end pipeline



- **Top (offline, do once):** fit HMM, solve MDP, build Q^*. VI and PI agree to 2.6×10^{-6}.
- **Bottom (each month):** new observation $\to$ updated belief $\to$ QMDP $\to$ weights $\to$ realised return.
- **Walk-forward** 2003-10 $\to$ 2026-04, monthly rebalance, 5 bps tx cost, 271 decisions.

QMDP rule, with a worked example

$$\boxed{\pi_{\mathrm{QMDP}}(\mathbf{b}) = \arg\max_{\mathbf{a}} \sum_{s} b(s)\, Q^*(s, \mathbf{a})}$$

Decision moment. After a VIX spike the belief filter returns $\mathbf{b} = (0.07, 0.93)$ (bear-leaning). At the *headline* $\gamma{=}2$, the Q^* table is genuinely regime-differentiated:

Action	$Q^*(\text{bull})$	$Q^*(\text{bear})$	$\mathbf{b}^\top Q^*(\cdot, \mathbf{a})$
100/0 stocks	0.183	0.024	0.035
60/40	0.180	0.028	0.038
0/100 bonds	0.174	0.032	**0.042**

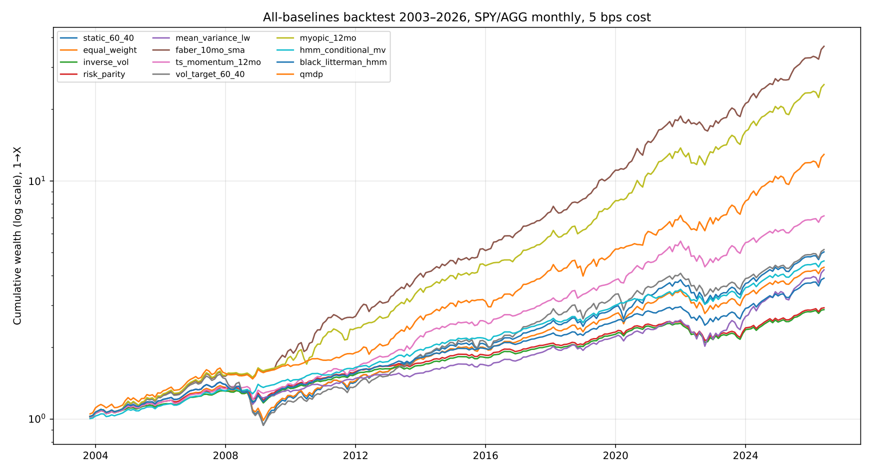
Bull column rewards stocks (argmax = 100/0); bear column rewards bonds (argmax = 0/100). The bear-leaning belief makes QMDP pick 0/100: a genuine regime-driven decision at the canonical $\gamma{=}2$.

Twelve-strategy horse race (2003–2026, 5 bps cost)

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Faber 10-mo SMA	**17.24%**	9.80%	**1.68**	**3.57**	−13.5%	**1.28**
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01
HMM-conditional MV	6.98%	5.76%	1.20	2.12	−13.5%	0.52
QMDP (γ=2)	11.96%	10.04%	**1.18**	2.13	**−14.2%**	0.84
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41
Black–Litterman + HMM	6.20%	6.28%	0.99	1.58	−17.8%	0.35
Inverse-vol	4.78%	5.24%	0.92	1.42	−16.8%	0.28
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23
Mean-variance + Ledoit-Wolf	6.58%	8.20%	0.82	1.32	−22.5%	0.29
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
Vol-target 60/40	7.50%	10.14%	0.77	1.12	−39.1%	0.19

Three takeaways. (1) QMDP finishes 4th of twelve (Sharpe 1.18), beating the static 60/40 (0.81) and every classical baseline at less than half the drawdown. (2) It ties HMM-conditional MV (1.20): two ways of using the same HMM signal land together. (3) Faber (1.68) still leads, so we are competitive with but do not overtake trend-following.

Equity curves: $1 invested 2003-10-31 (log scale)



Faber (top) leads. QMDP finishes comfortably above the static 60/40 with visibly shallower drawdowns in 2008 and 2020, because it rotates to bonds when the belief turns bear. Static 60/40 trails with the deepest trench (−34%).

Regime differentiation is robust across risk aversion

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	100/0	100/0	80/20	60/40
Bear (1)	0/100	0/100	0/100	0/100	0/100	0/100	0/100

The two rows differ in every column.

The bear regime is pinned at 100% bonds at every γ;
the bull regime only tempers toward 60/40 as γ rises.

Why it works: the bear regime's mean SPY return is *negative* (-0.22%/mo) while bonds stay positive, so even over a $\lambda=0.95$ horizon the optimal bear action is fully defensive. The regime label genuinely moves the policy at the canonical $\gamma=2$.

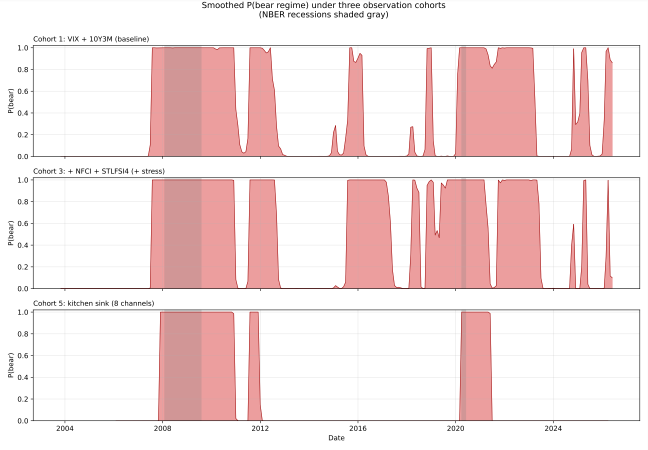
Robustness 1: every observation cohort differentiates

Cohort	Features	Bull	Bear	Differs?	BIC
1. Baseline	VIX, T10Y3M	100/0	0/100	✓	614
2. +Yield curve	+T10Y2Y	100/0	80/20	✓	641
3. +Stress	**+NFCI, +STLFSI4**	**100/0**	**0/100**	✓	**513**
4. +Macro	+NFCI, +UMCSENT, +ICSA	100/0	80/20	✓	1060
5. Kitchen sink	8 channels	100/0	0/100	✓	1069

Regime differentiation does *not* hinge on any one observation choice: even the two-channel baseline produces a fully defensive bear policy. Adding NFCI (Chicago Fed)
+ STLFSI4 (St. Louis Fed) attains the **lowest BIC** (513).

"More thermometers, sharper diagnosis," exactly as Guidolin–Timmermann predict.

Visual confirmation of Unlock 1



Top: cohort 1 (baseline, noisy false alarms). Middle: cohort 3 (+NFCI, +STLFSI4), with sharp bear spikes cleanly bracketing every NBER recession. Bottom: cohort 5 (kitchen sink, over-fits).

Robustness 2: walk-forward refit improves it further

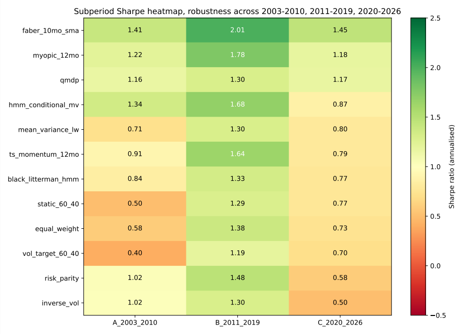
Variant	CAGR	Sharpe	Max DD	Calmar
Static 60/40 (benchmark)	7.40%	0.81	−34.2%	0.22
QMDP fixed (this pipeline)	9.87%	0.96	−34.2%	0.29
QMDP annual refit (5y win)	9.11%	0.85	−25.1%	0.36
QMDP quarterly refit	8.56%	0.83	−23.4%	0.37
QMDP expanding window	9.81%	**1.08**	**−16.5%**	**0.60**
HMM-MV expanding window	5.32%	0.95	−16.8%	0.32

Within this pipeline (fixed Sharpe 0.96), expanding-window refit (Nystrup et al. 2018) lifts

Sharpe 0.96 → **1.08** · Max DD −34% → **−16.5%** · Calmar 0.29 → **0.60**.

"Keep re-reading the casino's recent hands": current parameters react to new stress.

Robustness 3: QMDP is solid in every sub-period



- QMDP beats the static 60/40 in all three sub-periods: Sharpe **1.16** (A, 2003–10), **1.30** (B, 2011–19), **1.17** (C, 2020–26).
- Period A is the GFC era, exactly where a regime model should help, and QMDP delivers (1.16 vs. the static's 0.50).
- Faber still leads every column (1.41 / 2.01 / 1.45): we are consistently mid-pack and competitive, never the laggard.

The honest verdict: claim vs. non-claim

We DO claim	**We do NOT claim**
POMDP is the right *minimum* decision-theoretic tool for this problem.	POMDP asset allocation overtakes simple trend-following (Faber, 1.68).
At $\gamma=2$ QMDP beats the static 60/40 (Sharpe 1.18 vs. 0.81) and ties model-based MV (1.20).	QMDP is a production-ready strategy net of every real-world friction.
The belief state is genuinely recursive and regime-tracking; the QMDP rule is derived, not asserted.	Our specific Sharpe number is the point.
The regime-aware policy is robust across γ, observations, sub-periods, and refit.	A two-risky-asset universe is where regime overlays shine most.

A correctly-specified POMDP overlay is a competitive risk-adjusted allocator that beats the static benchmark. The framework and its robustness outlast any one sample's Sharpe number.

Three extensions in priority order

1. CVaR-penalised reward.
   Replace CRRA utility with $R = \mathbb{E}[r] - \kappa\,\mathrm{CVaR}_\alpha[r]$ to explicitly penalise bear-regime tail risk. One-line change in the reward function.

2. Point-based value iteration (PBVI).
   Maintain explicit α-vectors over a small set of reachable beliefs. Gives a tighter upper bound on QMDP. On a 1-D belief simplex the cost is negligible.

3. Off-policy MC + Double Q-learning.
   A genuinely model-free baseline that does *not* assume the QMDP transition matrix is correct. Cross-checks the model-based pipeline against learning from scratch.

Thank you.

Questions?

AI tools disclosure: Claude (Anthropic) helped generate diagrams and polish LaTeX math formatting.